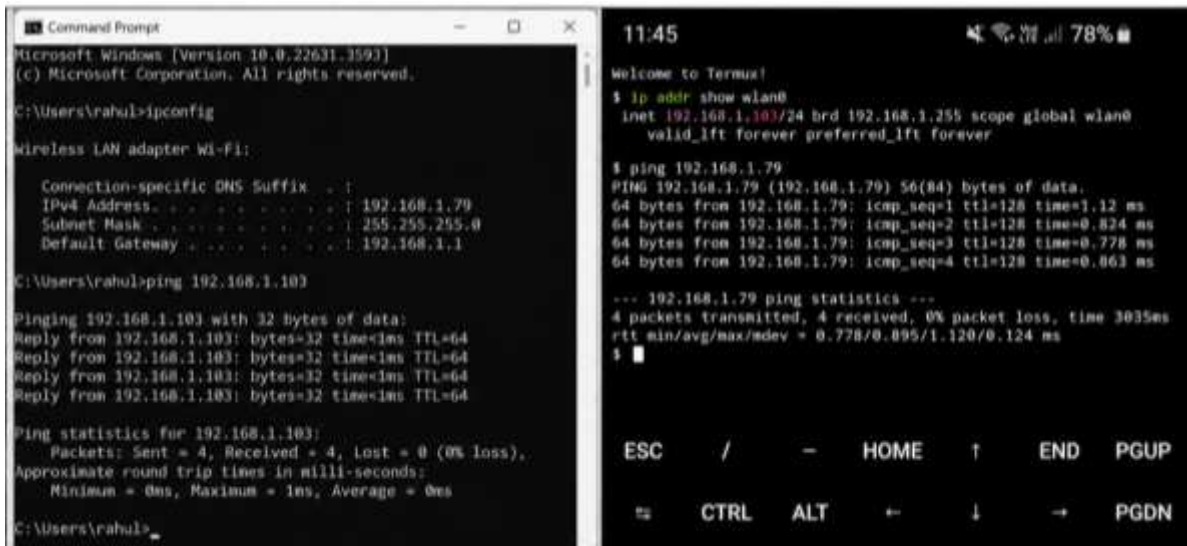


ASSIGNMENT 3.5

Q1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.



The image contains two side-by-side screenshots. The left screenshot is a Windows Command Prompt window showing the output of the 'ipconfig' command for a wireless LAN adapter, displaying an IPv4 address of 192.168.1.79 and a default gateway of 192.168.1.1. It also shows the output of a 'ping 192.168.1.103' command, indicating successful communication with 0% packet loss. The right screenshot is a Termux terminal on an Android phone showing the output of 'ip addr show wlan0' (displaying 192.168.1.103) and a 'ping 192.168.1.79' command, also showing successful communication with 0% packet loss.

```
Microsoft Windows [Version 10.0.22631.3593]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rahul>ipconfig

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : 
    IPv4 Address. . . . . : 192.168.1.79
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\rahul>ping 192.168.1.103

Pinging 192.168.1.103 with 32 bytes of data:
Reply from 192.168.1.103: bytes=32 time<1ms TTL=64
Reply from 192.168.1.103: bytes=32 time<1ms TTL=64
Reply from 192.168.1.103: bytes=32 time<1ms TTL=64
Reply from 192.168.1.103: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.1.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\rahul>
```

```
11:45
Welcome to Termux!

$ ip addr show wlan0
inet 192.168.1.103/24 brd 192.168.1.255 scope global wlan0
    valid_lft forever preferred_lft forever

$ ping 192.168.1.79
PING 192.168.1.79 (192.168.1.79) 56(84) bytes of data:
64 bytes from 192.168.1.79: icmp_seq=1 ttl=128 time=1.12 ms
64 bytes from 192.168.1.79: icmp_seq=2 ttl=128 time=0.824 ms
64 bytes from 192.168.1.79: icmp_seq=3 ttl=128 time=0.778 ms
64 bytes from 192.168.1.79: icmp_seq=4 ttl=128 time=0.863 ms

--- 192.168.1.79 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3035ms
rtt min/avg/max/mdev = 0.778/0.895/1.120/0.124 ms
$
```



Q2 Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:4829:400::]
over a maximum of 30 hops:

  1      4 ms      2 ms      2 ms      2401:4900:ae50:57bd::2a
  2      *        *        *        Request timed out.
  3     25 ms      *        *        2401:4900:0:13c9::1
  4     58 ms     58 ms     18 ms     2401:4900:0:af4::5
  5      *        25 ms      *        2401:4900:0:af5::1
  6     30 ms     21 ms     28 ms     2401:4900:4c:c409::25ab
  7     15 ms     22 ms     32 ms     2404:a800:2a00:20a::4a
  8     26 ms     18 ms     14 ms     2404:a800:2a00:20a::49
  9     40 ms     44 ms     48 ms     2404:a800::167
 10     35 ms     33 ms     38 ms     2001:4860:4829:400::

Trace complete.
```

I ran the `tracert www.youtube.com` command on my Windows device. The route took 10 hops to reach the destination. The trace was completed successfully.

Number of hops: 10.

Q3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

```
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Q4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

```
C:\Users\Admin>arp -a

Interface: 192.168.30.1 --- 0x7
    Internet Address      Physical Address          Type
    192.168.30.254        00-50-56-ff-9b-b2        dynamic
    192.168.30.255        ff-ff-ff-ff-ff-ff        static
    224.0.0.22            01-00-5e-00-00-16        static
    224.0.0.251           01-00-5e-00-00-fb        static
    224.0.0.252           01-00-5e-00-00-fc        static
    231.237.113.100       01-00-5e-6d-71-64        static
    239.192.152.143       01-00-5e-40-98-8f        static
    239.255.255.250       01-00-5e-7f-ff-fa        static
    255.255.255.255       ff-ff-ff-ff-ff-ff        static

Interface: 192.168.133.1 --- 0x10
    Internet Address      Physical Address          Type
    192.168.133.254        00-50-56-e9-17-a6        dynamic
    192.168.133.255        ff-ff-ff-ff-ff-ff        static
    224.0.0.22            01-00-5e-00-00-16        static
```

IP Address	Gadget/Device	Owner
10.41.171.31	Android Phone (Hotspot)	Admin
192.168.30.254	Virtual Network Adapter	System
192.168.133.254	Virtual Network Adapter	System

Q5 Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.
Hint: Include screenshots or command outputs if possible.

Network Configuration :

The network was configured using a wireless connection. A laptop was connected to the network through a Wi-Fi adapter. The system automatically obtained an IP address from the router using DHCP. After connecting to the network, the assigned IP address and network details were verified using standard Windows networking commands.

```
C:\Users\Admin>arp -a

Interface: 192.168.30.1 --- 0x7
    Internet Address      Physical Address      Type
    192.168.30.254        00-50-56-ff-9b-b2    dynamic
    192.168.30.255        ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251          01-00-5e-00-00-fb    static
    224.0.0.252          01-00-5e-00-00-fc    static
    231.237.113.100       01-00-5e-6d-71-64    static
    239.192.152.143       01-00-5e-40-98-8f    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.133.1 --- 0x10
    Internet Address      Physical Address      Type
    192.168.133.254       00-50-56-e9-17-a6    dynamic
    192.168.133.255       ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251          01-00-5e-00-00-fb    static
    224.0.0.252          01-00-5e-00-00-fc    static
    231.237.113.100       01-00-5e-6d-71-64    static
    239.192.152.143       01-00-5e-40-98-8f    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

Connectivity Testing Using Ping :

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Request timed out.
Request timed out.
```

Observation

The ping test failed and displayed "**Request timed out**" messages. This indicated that the destination device or network path was unavailable. The packet loss rate was 100%, confirming that communication could not be established.

Route Testing Using Traceroute :

```
C:\Users\Admin>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1    2 ms    2 ms    2 ms  10.41.171.31
  2   189 ms   202 ms   202 ms  192.168.17.10
  3    52 ms    41 ms    30 ms  192.168.16.33
  4    34 ms    21 ms    34 ms  192.168.19.3
  5    26 ms    32 ms    33 ms  192.168.19.33
  6    37 ms    48 ms    27 ms  192.168.252.42
  7    26 ms    17 ms    30 ms  125.20.216.10
  8    42 ms   100 ms    40 ms  125.20.216.9
  9    36 ms    40 ms    49 ms  116.119.73.209
 10   43 ms    73 ms    46 ms  72.14.212.48
 11   52 ms    69 ms    43 ms  142.251.225.67
 12   52 ms    35 ms    39 ms  192.178.86.247
 13   96 ms    41 ms    38 ms  dns.google [8.8.8.8]

Trace complete.
```

Issues:

1. **Request Timed Out Errors**
 - Occurred when a device was disconnected from the network.
 - Resulted in 100% packet loss during ping testing.
2. **Network Reachability Problems**
 - Communication failed when the target device was offline.
 - Connectivity was restored after reconnecting the device.
3. **Potential Causes**
 - Device disconnected from Wi-Fi.
 - Firewall blocking ICMP requests.
 - Temporary network interruption.